

Arizona State University
Department of Economics
Ph.D. Qualifying Exam

Macroeconomic Theory

June 3, 2022

Instructions: Read the questions carefully, show all your work, and explain all your answers. Indicate carefully if you introduce notation, or make assumptions when answering a question. Please make sure to allocate time to ALL three problems.

Good luck!

1. Consider a decision problem of an infinitely-lived agent who has initial capital endowment $k_0 > 0$ and can invest, from period 0 onwards, in one of the two projects, S (safe) or R (risky). The projects differ in two dimensions: the rates of return, A_S and A_R , and the probabilities of survival, γ_S and γ_R , with $\gamma_S > \gamma_R$. If the agent invests k_{t+1} in project $i = S, R$ in period t , the project pays off $A_i k_{t+1}$ in period $t+1$ with probability γ_i , which can be allocated between consumption c_{t+1} and further project investment. With probability $1 - \gamma_i$, the project fails, the agent's capital k_{t+1} disappears and (s)he receives a stream of consumption insurance benefits $b > 0$ for the rest of life.

Both projects are available in period 0, and start paying off (or fail) in period 1, so consumption in period 0 is $c_0 = k_0 - k_1$. The project choice can only be made in period 0, and cannot be adjusted thereafter. The agent cannot choose to abandon the project to receive benefit b (which can be granted only conditional on verifiable project failure and loss of capital). The agent's life-time utility is given by

$$\mathbb{E} \sum_{t=0}^{\infty} \beta^t \ln(c_t),$$

where $\beta \in (0, 1)$.

- (a) Set up the agent's decision problem recursively.
- (b) Show that the agent's project choice in period 0 is characterized by a cutoff rule. Namely, there exists $\bar{k}_0 \geq 0$ such that the agent chooses project S (the one with higher survival probability) if and only if $k_0 \geq \bar{k}_0$. Provide a brief intuitive interpretation for this result.
- (c) Now suppose that $A_S > A_R$. Your classmate claims that because project S has higher return and is more likely to survive, the agent would choose higher initial consumption if (s)he were to operate project S compared to operating project R . Develop a formal argument to either support this claim or prove that it is wrong.
- (d) Consider a modification of the model in which the agent can choose between (A_S, γ_S) and (A_R, γ_R) in every period, until the project fails. Show how this modification affects the setup of the agent's decision problem, and derive the conditions on (A_S, γ_S) and (A_R, γ_R) under which the agent's optimal project choice remains constant for as long as the project is alive.

2. Consider the following version of the overlapping generations model with capital. Time is discrete ($t = 0, 1, 2, \dots$). Individuals live for two periods, and work in the market in only the first period of their lives. Competitive firms have access to a Cobb-Douglas technology that uses capital (K) and labor services (L). For simplicity, assume that there is no growth in population or in labor efficiency.

In this economy, individuals value regular consumption c , and home produced goods, h (home cooking, cleaning, gardening, etc.) Home production requires only labor time, l . Individuals produce home goods when young and old according to $h_{i,t} = G(l_{i,t})$, $i = 1, 2$. The function G is strictly increasing, concave and differentiable, and satisfies $G(0) = 0$. Home goods are perishable and storage is not possible. The total endowment of time available for each individual is equal to 1.

An individual born at date t of any type chooses consumption levels $c_{1,t}$, $c_{2,t+1}$, home goods $h_{1,t}$, $h_{2,t+1}$ and savings s_t , to maximize

$$U(.) = \ln(c_{1,t}) + \gamma \ln(h_{1,t}) + \beta [\ln(c_{2,t+1}) + \gamma \ln(h_{2,t+1})],$$

where $\beta > 0$.

- (a) Formulate the problem of a young individual in period t . Derive the first order-conditions that characterize optimal choices. Explain.
- (b) Assume from now on that there are two types of individuals born into this economy. Type- L individuals have a valuation of home goods γ_L , while type- H individuals have γ_H , with $\gamma_L < \gamma_H$. Type- L individuals constitute a fraction π of each cohort. Furthermore, the home technology is given by $G(l) = l^\lambda$, $\lambda \in (0, 1)$.

Derive the law of motion for the capital-to-labor ratio. Show that there is a unique steady state in this economy, in which capital-to-labor ratio is strictly positive.

- (c) Consider two economies, A and B . These economies differ in the relative fractions of type- L individuals, with $\pi_A > \pi_B$. Given this observation, economist X states that in steady state, GDP per worker will be higher in A . Shen then states that because type- L individuals save more than type- H individuals, the rate of return on capital will be lower in economy A . Is she right? Explain carefully.

3. This problem asks you to analyze a version of the McCall search model with temporary unemployment compensation. Each period an unemployed worker draws one offer to work at wage w . Wages are i.i.d. draws from the c.d.f. F , where $F(0) = 0$ and $F(\bar{w}) = 1$. The worker seeks to maximize $\mathbb{E} \sum_{t=0}^{\infty} \beta^t y_t$, where y_t is the sum of the worker's wage and unemployment compensation, if any, and $\beta \in (0, 1)$. The worker is entitled to unemployment compensation in the amount $b > 0$ only during the first period that she is unemployed. After one period on unemployment compensation, the worker receives none.

An employed worker can quit the job. If she does, she is again entitled to unemployment compensation for the first period of the unemployment spell. (To requalify for unemployment compensation, the worker must find a job and work for at least one period.) The timing is as follows. At the very beginning of a period, a worker who was employed in the previous period must decide whether or not to quit. The decision is irreversible; that is, a quitter cannot return to an old job. If the worker quits, she receives the unemployment benefit b this period (and this period only) and then draws a new wage offer next period as described above. All workers accepting a job start working at that job next period.

- (a) Write the Bellman equations for this problem. *[Hint: At the beginning of a period, let $v_e(w)$ denote the value of a worker who was employed in the previous period with wage w (before any wage draw in the current period). Let $v_1(w)$ be the value of an unemployed worker who has drawn wage offer w and who is entitled to unemployment compensation if she rejects the offer. Similarly, let $v_+(w)$ be the value of an unemployed worker who has drawn wage offer w but who is not eligible for unemployment compensation.]*
- (b) Prove that the optimal strategies of the employed and unemployed worker are of the reservation wage form. Denote the reservation wages of the employed, unemployed for one period, and unemployed for 2+ periods by w_e^* , w_1^* , and w_+^* , respectively.
- (c) How do w_e^* , w_1^* , and w_+^* compare to each other? Provide both proofs of your statements as well as intuition. Show that the model predicts that workers with (weakly) longer past unemployment spells are more likely to quit their jobs.
- (d) Write down an expression for the difference $w_1^* - w_+^*$ and interpret it intuitively.